9			**DR** Gradient of normal = 2 $\Rightarrow$ gradient of tangent = -0.5	**M1**	**3.1a**	$\frac{-y}{x+2y}=-\frac{1}{2}$	**Or M1** for correct expression for x in terms of y and attempt to differentiate
			$2y\frac{dy}{dx}+y+x\frac{dy}{dx}=0$	**M1** **A1**	**1.1** **1.1**	Attempt at implicit differentiation with one algebraic term differentiated correctly	
			eg $-y+y-0.5x=0\Rightarrow x=0$	**A1**	**2.2a**	Correct implicit expression	For one step solving their equation in x and/or y with original equation
			$y^2=8$	**M1**	**1.1**	Implied by at least one y-value	
			$\left(0,\ \sqrt{8}\right)$ and $\left(0,\ -\sqrt{8}\right)$	**A1**	**1.1**		

			Alternative method $\frac{dx}{dy} = \frac{-8-y^2}{y^2}$ oe $\frac{8+y^2}{y^2} = 2$ oe $y^2 = 8$ $\left(0, \quad \sqrt{8}\right)$ and $\left(0, \quad -\sqrt{8}\right)$	**M1** **M1** **A1** **M1** **M1** **A1** **[6]**		Rearranging to $x = \frac{8-y^2}{y}$ and attempting differentiation Differentiation by dividing first or quotient rule – allow one error Correct differentiation Using negative reciprocal Implied by at least one y-value	

			$f'(x) = \frac{(1+e^x)e^x - e^x e^x}{}$			Use of quotient rule or product rule	